

Formulae

① $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

A diagram showing the formula $S = \frac{D}{T}$ enclosed in a rectangular box. To the right of the box, a dashed green arrow points from the letter 'D' to the letter 'm', and another dashed green arrow points from the letter 'T' to the letter 's'. A solid green arrow originates from the left side of the box and points down to the unit '(m/s)' written in green.

$$S = \frac{D}{T}$$

(m/s)

Q] A car travels 100m in 10 sec. Find speed of car. $[km/hr]$ ✓✓

Solⁿ: ① $D = 100m$

$$T = 10s$$

$$S = \frac{D}{T} = \frac{100m}{10s} = 10 (m/s)$$

$$② \quad S = 10 \frac{m}{s} \rightarrow ? km/hr$$

$$S = 10 \frac{\text{m}}{\text{s}}$$

$$= 10 \times \frac{(\text{km})}{\text{ks}}$$

$$= \frac{10}{\text{k}} \frac{3600 (\text{km})}{3600 \text{ s}}$$

$$S = \frac{3600 \times 10}{\text{k}} \left(\frac{\text{km}}{\text{hr}} \right)$$

$$\left(\frac{\text{km}}{\text{hr}} \right)$$

$$1 \text{ hr} = 60 \text{ min}$$

$$= 60 \times 1 \text{ min}$$

$$= 60 \times 60 \text{ s}$$

$$1 \text{ hr} = 3600 \text{ s}$$

$$S = \frac{3600 \times 10}{k} \left(\frac{\text{km}}{\text{hr}} \right)$$

$$= \frac{\cancel{3600}^{18}}{\cancel{1000}_5} \times 10 \left(\frac{\text{km}}{\text{hr}} \right)$$

$$S = 36 \text{ km/hr}$$

Conceptual

$$(Q) \frac{m}{s} \longrightarrow (Q) \times \frac{18}{5} \frac{km}{hr}$$

$$\left(P \times \frac{5}{18}\right) \frac{m}{s} \longleftarrow (P) \frac{km}{hr}$$

Q] A car travels 200m with speed of 72 km/hr. How much time (in sec) it takes to travel the distance?

Solⁿ :

① $D = 200 \text{ m}$

$S = 72 \text{ km/hr}$

② $S = \frac{D}{T} \rightarrow T = \frac{D}{S} = \frac{200 \text{ m}}{72 \text{ km/hr}}$

$T = \frac{200 \cancel{\text{m}}}{72 \times \frac{5}{18} \cancel{\text{m/s}}} = 10 \text{ s} //$

Q] I travel 3 distances in the ratio 1:2:3 at speed of 20m/s, 30m/s, 40m/s. If the total time taken is 23 minutes. Find the total distance covered?

① Distance $\Rightarrow d_1 : d_2 : d_3 = 1 : 2 : 3 = k : 2k : 3k$

$$\Rightarrow d_1 = k, d_2 = 2k, d_3 = 3k$$

② Speed $\Rightarrow S_1 = 20 \text{ m/s}$

$$S_2 = 30 \text{ m/s}$$

$$S_3 = 40 \text{ m/s}$$

$$\begin{aligned}\textcircled{3} \quad \text{total distance} &= d_1 + d_2 + d_3 \\ &= k + 2k + 3k \\ &= 6k\end{aligned}$$

$$S = D/T$$

$$\boxed{T = D/S}$$

$$\textcircled{4} \quad T = 23 \text{ min} = 23 \times 60 \text{ s}$$

$$\textcircled{t_1} + \textcircled{t_2} + \textcircled{t_3} = 1380$$

$$\frac{D_1}{S_1} + \frac{D_2}{S_2} + \frac{D_3}{S_3} = 1380$$

$$\frac{D_1}{S_1} + \frac{D_2}{S_2} + \frac{D_3}{S_3} = 1380$$

$\frac{m}{s}$

$$\frac{k}{20} + \frac{2k}{30} + \frac{3k}{40} = 1380 \text{ s}$$

$$k = 7200$$

$$\text{Total distance} = 6k = 6 \times 7200 = 43200 \text{ m}$$

$$= 43200 \text{ m}$$

$$= 43200 \times \frac{\text{km}}{1000}$$

$$= \frac{43200}{1000} \text{ km}$$

$$= 43.2 \text{ km} //$$

Q] I travel 3 stretches of time in the ratio 1:2:3 at speed of 20m/s, 30m/s, 40m/s and cover a total distance of 100 km. Find the total time for which I am travelling?

Solⁿ : ① $t_1 = \underline{k}, t_2 = \underline{2k}, t_3 = \underline{3k}$

② $s_1 = 20 \text{ m/s}, s_2 = 30 \text{ m/s}, s_3 = 40 \text{ m/s}$

③ $d_1 + d_2 + d_3 = 100 \text{ km}$

④ $T = t_1 + t_2 + t_3$
 $= 6k$

$$d_1 + d_2 + d_3 = 100 \text{ km}$$

$$S = \frac{D}{T}$$

$$S_1 t_1 + S_2 t_2 + S_3 t_3 = 100 \times 10^3 \text{ m} \Rightarrow D = S T$$

$$20(k) + 30(2k) + 40(3k) = 10^5 \text{ m}$$

$$k = 500$$

$$T = 6k = 6 \times 500 = 3000 \text{ s}$$

Q] If a person travels from his home 6 km/hr slower he reaches his office 8 min late but if he travels 4 km/hr faster he reaches 4.5 min earlier. The distance between his home and office is _____

(5★)

① Distance = D

②

$$S = \frac{D}{T}$$

[originally]

③

$$T' = T + 8 \Rightarrow T' - T = 8 \text{ min}$$

$$\frac{D}{S'} - \frac{D}{S} = 8 \text{ min}$$

$$\frac{D}{S'} - \frac{D}{S} = 8 \text{ min}$$

$$\frac{D}{S-6} - \frac{D}{S} = 8 \text{ min}$$

$$D \left[\frac{1}{S-6} - \frac{1}{S} \right] = 8 \times \frac{1}{60} \text{ hr}$$

$$\Rightarrow S(S-6) = 45D \quad \text{--- 1}$$

④

$$T_1 = T - 4.5$$

$$4.5 = T - T_1$$

$$4.5 \text{ min} = T - T_1$$

$$T = \frac{D}{S}$$

$$4.5 \times \frac{1}{60} \text{ hr} = \frac{D}{S} - \frac{D}{S_1}$$

$$\frac{4.5}{60} = D \left[\frac{1}{S} - \frac{1}{S+4} \right]$$

$$\Rightarrow 3S(S+4) = 160D \quad \text{--- 2}$$

$$\cdot 1 \oplus 2 \Rightarrow \frac{s-6}{3(s+4)} = \frac{9}{32}$$

$$s = 60$$

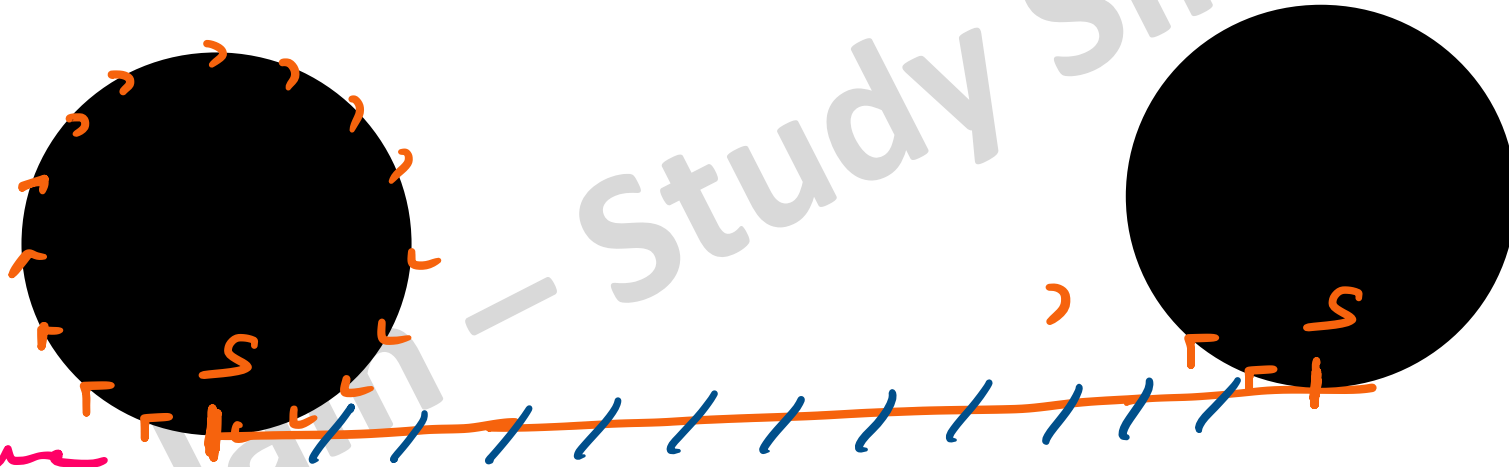
$$3s(s+4) = 160 \quad \text{D}$$

$$\Rightarrow D = \frac{3s(s+4)}{160} = 72 \text{ km} //$$

Q] A wheel of radius 2.1m of a vehicle makes 75 revolution in 1 min. What is the speed of the vehicle? [km/h]

Solⁿ :

circle
↓
Circumference
= $2\pi r$



Distance = 1 revolution

$$= 2\pi r$$

$$= 2\pi(2.1\text{ m}) = 13.18\text{ m}$$

1 min ✓

75 revolution

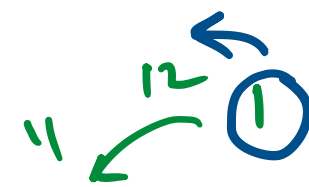
$$S = \text{km/hr}$$

$$S = \frac{D}{T} = \frac{75 \times (13.18 \text{ m})}{60 \text{ sec}} = 16.475 \text{ m/s}$$

$$16.475 \times \frac{18}{5} \frac{\text{km}}{\text{hr}}$$

$$= 59.4 \text{ km/hr} //$$

Q] A man cycles with a speed of 10km/hr and reaches his office at 1pm. However, when he cycles with a speed of 15km/hr he reaches his office at 11am. At what speed should he cycle so that he reaches his office at 12 noon.



① $S = 10 \text{ km/hr}$, t , $D = [\text{Same}]$

② $S = 15 \text{ km/hr}$, $t-2$, D

③ $S = ?$, $t-1$, D

$$S = D/T$$

$$\Rightarrow \boxed{D = ST}$$

$$\Rightarrow D = \text{same} = \text{constant}$$

$$\Rightarrow S_1 T_1 = S_2 T_2$$

$$10(t) = 15(t-2)$$

$$t = 6 \text{ hr}$$

$$S = ? , \textcircled{t-1}, D \checkmark$$

$$S = \frac{D \checkmark}{T} = \frac{60 \text{ km}}{(6-1) \text{ hr}} = 12 \text{ km/hr} //$$

$$S = 10 \text{ km/hr} , \textcircled{t} , D = [\text{Same}]$$

$$D = ST = 10 \text{ km/hr} \times 6 \text{ hr} = 60 \text{ km}$$

$$\text{Speed} = \frac{\text{Dist}}{\text{Time}}$$

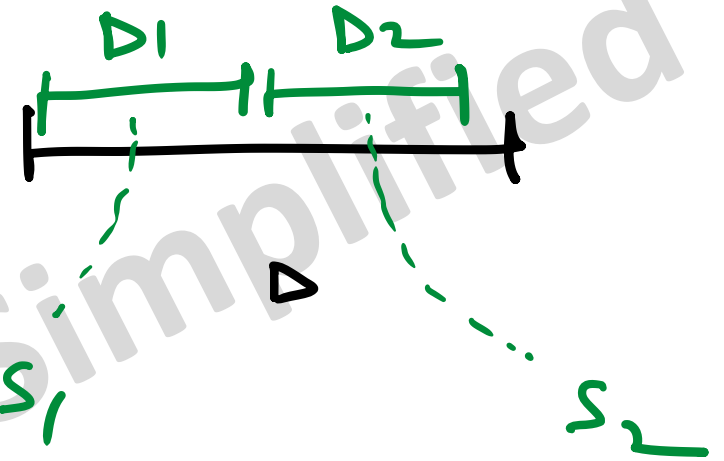
$$\text{Average Speed} = \frac{\text{Total dist}}{\text{Total time}}$$

$$(i) \quad D = D_1 + D_2$$

$$\text{Avg speed} = \frac{\text{Total Dis}}{\text{total time}}$$

$$= \frac{D_1 + D_2}{t_1 + t_2}$$

$$\text{Avg speed} = \frac{D_1 + D_2}{\frac{D_1}{S_1} + \frac{D_2}{S_2}}$$



$$S = D/T$$
$$T = D/S$$

Q] A car covers first 400m with the speed of 72km/hr and next 300m with the speed of 36km/hr. Find out the average speed of the car (m/s).

$$T = D/S$$

$$\begin{aligned} \textcircled{1} \quad \text{Avg Speed} &= \frac{DT}{T_T} = \frac{D_1 + D_2}{T_1 + T_2} = \frac{700m}{\frac{400}{20} + \frac{300}{10}} \\ &= 14 \text{ m/s} \end{aligned}$$

$\textcircled{2}$



$$72 \text{ km/hr}$$

$$= 72 \times \frac{5}{18} = 20 \text{ m/s}$$

$$36 \text{ km/hr} = 10 \text{ m/s}$$

Wrmg :

$$\text{Avg } S = \frac{S_1 + S_2}{2}$$

$$= \frac{72 + 36}{2}$$

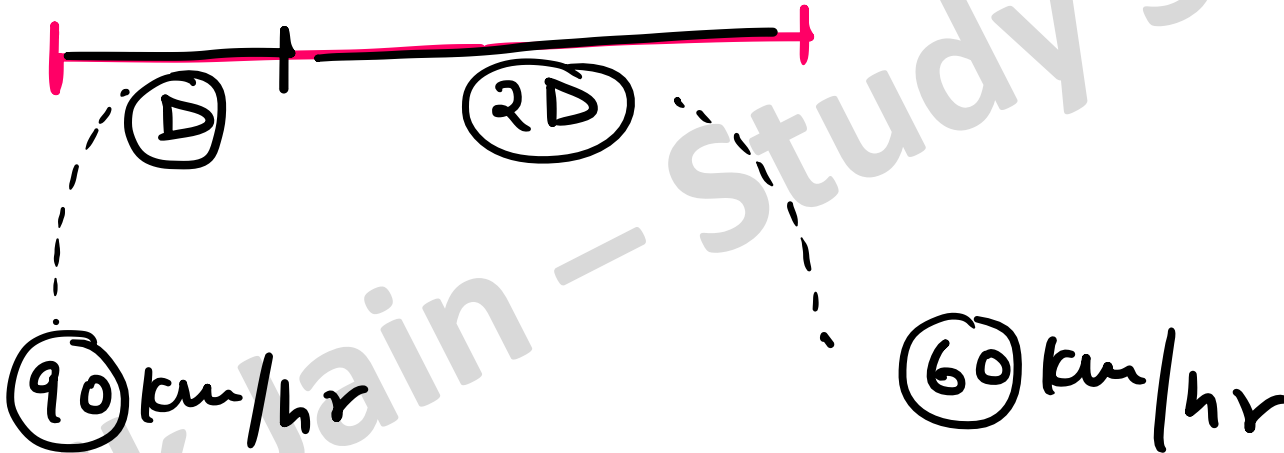
$$= 54 \text{ km/h} = 15 \text{ m/s}$$

Q] A train covers a distance with speed of 90 km/hr and then double of the previous distance with the speed of 60 km/hr . Find out its average speed

$$S = D/T$$

$$T = D/S$$

①



②

$$\text{Avg} = \frac{D + 2D}{T_1 + T_2} = \frac{3D}{\frac{D}{90} + \frac{2D}{60}} = 67.5\text{ km/hr}$$

Q] A car travels 8km in the first quarter of an hr, 6km in the second quarter and 16km in the third quarter. The average speed of the car in km per hr over the entire journey is

- (A) 30
- (B) 36
- ☒ (C) 40
- (D) 24

[GATE 2013 EE/EC/IN]

$$\frac{1}{4} \times 1 \text{ hr} \\ = \frac{1}{4} \text{ hr}$$

$$\text{Avg Speed} = \frac{D_T}{T_T}$$

$$= \frac{8 + 6 + 16}{\frac{1}{4} + \frac{1}{4} + \frac{1}{4}}$$

$$\text{Ans} = \frac{8 + 6 + 16}{\cancel{T_1} + \cancel{T_2} + \cancel{T_3}}$$

$\frac{1}{4} \quad \frac{1}{4} \quad \frac{1}{4}$

$$= \frac{30}{3/4} \text{ km/hr}$$

$$= 40 \text{ km/hr} //$$

Q] A tourist covers half of his journey by train at 60km/hr, half of the remainder by bus at 30km/hr and the rest by cycle at 10km/hr. The average speed of the tourist in km/hr during his entire journey.

$$S = D/T$$

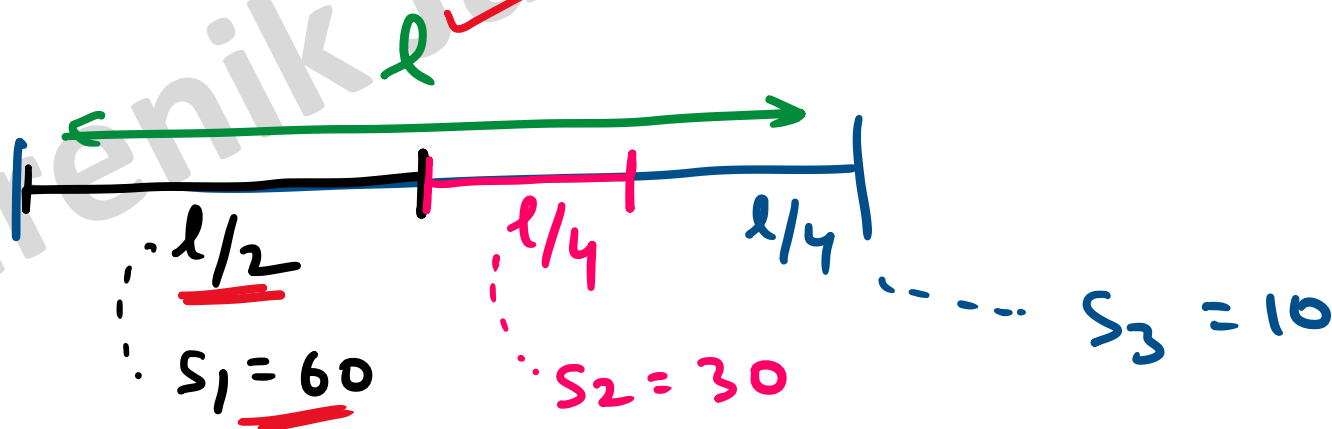
(A) 36

(B) 30

✓ (C) 24

(D) 18

[GATE 2013 CS/ME]



$$\textcircled{1} \text{ Avg } S = \frac{D_T}{T_T}$$

$$= \frac{l}{\frac{\frac{l}{2}}{60} + \frac{\frac{l}{4}}{30} + \frac{\frac{l}{4}}{10}}$$
$$= 24 \text{ km/hr}$$

Q] Budhan covers a distance of 19km in 2 hr by cycling one fourth of the time and walking the rest. The next day he cycles (at the same speed as before) for half the time and walks the rest (at the same speed as before) and covers 26km in 2 hr. The speed in km/hr at which Budhan walks is

$$S = D/T$$

$$D = ST$$

(A) 1

~~(B) 4~~

(C) 5

(D) 6

[GATE 2017 CE]

~~$$\textcircled{1} 2 \times 19 = 2 \times C \left(\frac{1}{4} \cdot 2 \right) + 2 \times W \left(\frac{3}{4} \cdot 2 \right)$$~~

~~$$\textcircled{2} 26 = C \left(\frac{1}{2} \cdot 2 \right) + W \left(\frac{1}{2} \cdot 2 \right)$$~~

$$W = 6 \text{ km/hr} //$$